Preterm Birth Raises Group-Level Cognitive Risk, Not a Sentence

A 2024 meta-analysis finds preterm birth associated with lower global cognitive scores (SMD about −0.57); the difference is a population risk, concentrated among the most immature and those with pulmonary morbidity, not an individual forecast.

GRADE: MODERATE-GROUP; LOW-INDIVIDUAL | STATUS: DRAFT | DOMAIN: BIRTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Being born before term exposes an infant to extrauterine life before some systems are ready. Across decades of standardized testing, children born preterm score lower, on average, than term-born peers on global cognitive measures. The gap is larger with more extreme immaturity and with neonatal pulmonary morbidity. Contemporary neonatal care is associated with early improvements that, in this synthesis, are not clearly retained after about age 2–3.

EVIDENCE AND CAUSAL ASSESSMENT

Burstein, Aryeh & Geva (2024): 161 studies, 39,799 children; global cognition SMD −0.57 (95% CI −0.63 to −0.52), about 7–9 developmental/IQ points. Observational comparison of preterm vs term; not an experiment that assigns gestational age. Examiner blinding was associated with poorer preterm scores, suggesting possible “compassion bias” when testers know history.

EFFECT SIZE AND PERSISTENCE

SMD ≈ −0.57 is an average. Moderate-to-late preterm cohorts do not show the same later widening as births before 32 weeks in this analysis. Early care-related gains that fade after toddlerhood are a persistence finding, not a reason to treat early scores as destiny.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (clinical cohort meta-analysis of gestational age, not an observational parenting phenotype).

Survival/referral bias; historical treatment change; confounding by indication; attrition. Corrected age matters in early assessment (clinical follow-up guidance such as NICE NG72 is adjacent policy, not this card’s identification).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Prematurity is not a developmental sentence. A group difference is not a child’s worth, personality, or a parental verdict.

WHAT ACTION IS REASONABLY JUSTIFIED

Risk-sensitive follow-up and family support. Using gestational age as a label that replaces the child is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Burstein, Aryeh & Geva (2024), Developmental Psychology (PMID 39480317)